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KINETIC AND KINEMATIC GAIT ANALYSIS IN DOGS

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Gait analysis can accurately assess normal and abnormal gait, identify characteristic features of specific gait abnormalities, and quantify gait so that numeric comparisons can be made between different populations.[2, 19] Kinesiology is the science of animal motion and includes kinetics (the forces that affect motion) and kinematics (the temporal and geometric characteristics of motion).[2] Kinetic gait analysis can be performed using a force plate to obtain a noninvasive, objective, and quantitative assessment of the forces occurring between the foot and the surface of the plate during the stance phase of the stride (ground-reaction forces).[2] Data relative to the swing phase of the stride are not measured. Successive strides during locomotion cannot be measured by a single force plate; however, a series of plates or a force plate built into a treadmill can be used to evaluate consecutive strides.[4, 21] Kinematic gait analysis is performed using a series of cameras and retroreflective targets placed on the dog's skin over specific anatomic landmarks.[1, 12] A computer measures the flexion and extension movements of joints and the temporal and distance variables of the dog's gait as it passes through a three-dimensional testing area. This method provides information relative to the stance and swing phase of the dog's stride. In most cases, kinematic analysis is combined with force-plate analysis so that dynamic three-dimensional characteristics of limb motion can be correlated with ground-reaction force measurements.[12]

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FORCE-PLATE ANALYSIS

Equipment

A force plate consists of sensing elements covered by a top plate. The force plate must be mounted firmly to a rigid level surface that does not yield or shift when force is applied. Ideally, the platform is attached to a heavy metal mounting plate, which is secured to a concrete foundation. The plate may be recessed into a floor, or a walkway can be constructed around the force plate level with the surface of the top plate. The walkway must not touch the force plate or it may interfere with data collection. It is important that the plate does not shift when force is applied to the top plate, or erroneous results may be obtained. The top plate is usually covered to provide a nonslip surface and to protect it from damage; however, the covering must not absorb impacting forces or significantly change the distance from the plate to the sensors. Force plates are available in numerous sizes. If the plate is small, the subject may not contact the plate properly during each pass. If the plate is large, the subject may strike the plate with more than one foot simultaneously during each pass and invalidate the trial.

When a subject steps on the force plate, the magnitude of the force is measured by deflection of the sensing elements within the plate.[2] The displacement of the sensing elements is proportional to the force applied to the plate and creates an electric signal, which is amplified and recorded by computer. Three orthogonal ground-reaction forces are measured: mediolateral (Fx), craniocaudal or braking/propulsion (Fy), and vertical (Fz) (Fig. 1).[2] Most force-plate systems can trigger data collection to begin when a minimum force is applied to the plate or when a

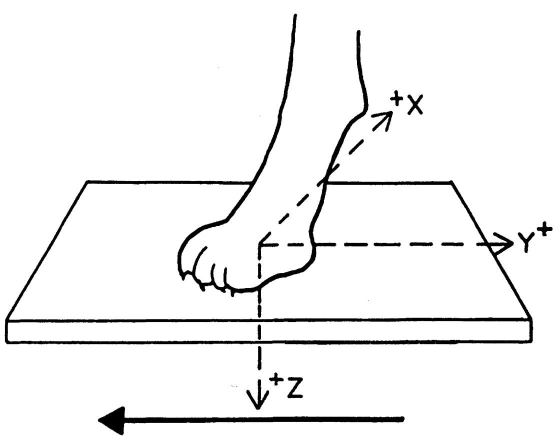

Figure 1. The three orthogonal ground-reaction forces measured during force-plate analysis. Solid arrow indicates forward progression. X = mediolateral force; Y = craniocaudal force; Z = vertical force.

minimum voltage is generated by the sensors. The system can thus be set to collect data beginning with the initial forelimb contact. Alternatively, collection can be triggered by the operator. The sampling frequency (measured in hertz as number of samples collected per second) and the sampling time (duration in seconds) can be adjusted and determine the amount of data collected. Collecting data at higher frequencies allows detection of minute rapid changes in gait; however, significantly more computer memory is required to store the data. The time of data collection depends on the velocity of the subject and should be long enough to include the portion of the stride being evaluated. In most cases, data are collected for the entire stance phase of the ipsilateral forelimbs and hind limbs. Force data are collected at the designated frequency, and a force-versus-time curve is generated by a computer. The force-versus-time curve is a graphic representation of the ground-reaction forces generated during the stance phase of the stride. Force is generally reported in newtons or kilograms of force; time is reported as seconds of stance time. Force plates can also measure vertical torque, center of pressure, and coefficient of friction, although these parameters are not commonly used in canine gait analysis.

A force plate may have strain gauges as the sensing elements.[2] When the top plate is loaded, deflection or bending of the metal foil gauge produces an electric output (microstrain) The material within the strain gauge changes electric resistance in proportion to the deformation. The strain gauges are connected to an electric circuit, which measures the change in resistance and determines the force applied to the plate. Strain gauge systems are durable and quite sensitive. Newer force plates have eliminated the "cross-talk" between strain gauges so that the three orthogonal forces can be accurately and independently measured. The strain elements undergo repeated mechanical deflections with use; thus, periodic adjustments of the bridge excitation voltage and recalibration of the system are typically necessary. Strain gauge force plates are able to measure forces within a specific range without recalibration; however, applying a force that overloads the strain gauge can cause permanent damage.

Piezoelectric force-plate systems rely on the inherent material properties of quartz crystals.[2] Quartz crystals respond to changes in shape by generating an electric charge (piezoelectric effect) and can be oriented to respond to external forces. Quartz crystals are placed in sensors within the force plate. When load is applied, crystal deformation produces voltage proportional to the applied force. The electric output is collected by electrodes attached to the surface of the crystals, amplified, and recorded by computer for display and analysis. Data from sensing elements in the four corners of the plate are integrated to determine the orthogonal ground reaction forces applied to the plate even if the force is not applied to the exact center of the plate. Piezoelectric systems are highly sensitive over a large measuring range. The same plate has been used to accurately measure the heartbeat of a standing subject (ballistocardiography) and impact forces from automobile crashes. Because piezoelectric systems rely on crystal deformation to generate an electric output, they can be extremely rigid, which allows for a higher natural frequency, a larger measuring range, and a durable plate. Piezoelectric force plates do not require periodic adjustments or recalibration, do not require an external electric energy source to obtain an output signal, and are not damaged by overloading.

Collecting Force-Plate Data

During data collection, the dog is led across the force plate by a handler. The dog must cross the plate in a consistent manner and the handler must not interfere in anyway while the dog is in contact with the plate. Pulling on the leash or minor changes in the dog's body position such as lowering or raising the head affect the ground-reaction force measurements obtained. The plate should be inconspicuously mounted so that the dog does not shy away or alter its gait when approaching the plate. Each pass across the plate is evaluated by an observer to confirm foot strikes and gait and to ensure that there is no interference with the stride. Video recording of each pass allows reevaluation and documentation of valid trials. A valid trial generally consists of distinct ipsilateral forefoot and hind foot strikes. If other feet contact the plate, or if more than one foot contacts the plate simultaneously, the forces applied by the individual limbs cannot be distinguished, and the trial is invalid. Repeated passes across the plate are usually required to obtain a sufficient number of valid trials. For each valid trial, orthogonal ground reaction forces in the forelimbs and hind limbs are measured and recorded. In most situations, variation is minimized by acquiring numerous valid trials and averaging the results to determine a representative gait pattern.

During each trial, the dog's forward velocity is measured. This can be achieved using a millisecond timer and two photoelectric switches.[26] The timer is started when the dog passes through one photoelectric beam and is stopped when the dog passes through the second beam. The known distance between the photoelectric switches is used to calculate the subject's velocity. Care must be taken to ensure that the dog consistently triggers the timer and that a constant speed is maintained across the plate during each trial. Acceleration or deceleration affects force values and can be difficult to control. Use of accelerometers may help to minimize variation during data collection.[5] Cinematography can also be used to determine the dog's velocity.

Numerous factors can affect the generation of ground-reaction forces during gait analysis.[20, 30] Studies evaluating the effect of velocity variation on ground-reaction forces and stance times in dogs at the walk and trot found that peak vertical forces increased as velocity increased.[24, 30, 31] Stance times decreased as velocity increased.[24–26, 29, 31] These findings demonstrate the importance of controlling the subject's speed during gait analysis. Only trials collected within a specific velocity range should be considered valid. If velocity is not limited to a narrow range, it is unclear if changes in ground-reaction force measurements are a result of gait differences or velocity variation. This is important when serially evaluating individual dogs (e.g., comparing pretreatment and posttreatment data) and when comparing data collected from different populations of dogs. Other factors known to cause variance in force-plate data include inherent subject variation, variation in trial repetitions for each subject, interday variation, handler differences, force-plate sensitivity, morphometric differences between subjects, and limb acceleration and deceleration.[2, 6, 20, 32] These potential causes of variance should be controlled carefully during data collection. When such factors are controlled by consistent data collection methods, the variance in force-plate data is low, providing an accurate, and repeatable assessment of limb function and allowing reliable comparisons between dogs and comparisons of clinical cases with known research data.

Evaluation of Force-Plate Data

Ground-reaction force data can be presented graphically as force-versus-time curves (Fig. 2). Mathematic transformation may be used to analyze the curves and identify subtle gait changes indicative of subclinical lameness.[19, 26, 36] Fourier transformation decomposes the force-versus-

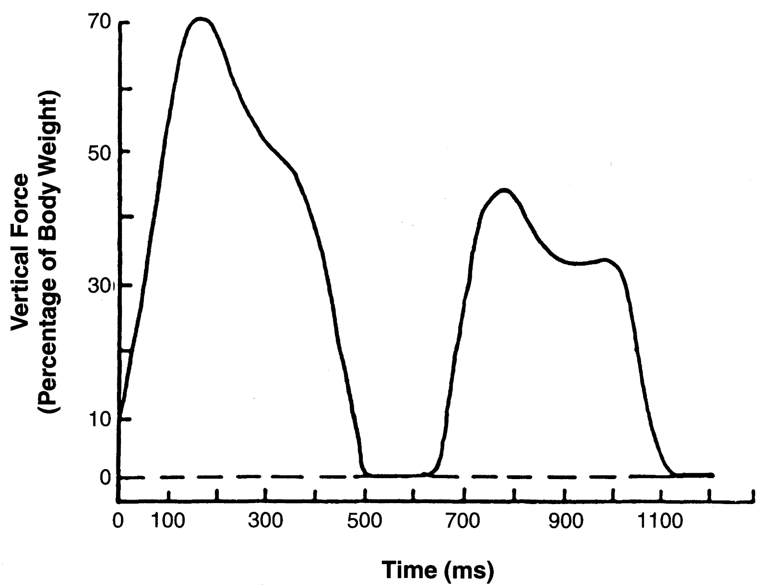

Figure 2. A force-versus-time curve from a normal dog at walk. The first peak is the forelimb stance phase; the second peak is the hind limb stance phase.

time curves into sine and cosine components, allowing mathematic reconstruction of the curve and making comparisons of graphic curves convenient and reliable.[36] These methods allow evaluation of the entire force-versus-time curve (or specified portions of the curve) but require complex mathematic and statistical procedures.

Force plate data may also be presented numerically. Selected peaks and slopes are chosen and analyzed for each of the three orthogonal ground reaction forces. Frequently evaluated parameters include peak (maximum) force, average force, impulse (force over time), rate of loading, and rate of unloading. Vertical force is the largest of the orthogonal forces and is the one most commonly evaluated for gait analysis in animals. Mediolateral and craniocaudal (braking/propulsion) forces are smaller in magnitude and may be more significantly affected by external factors. Force data are often normalized with respect to the dog's body weight, because evaluation of force as a percentage of body weight eliminates variation caused by changes in the weight of subjects and allows comparisons to be made between dogs of different weights. The mean values of several trials are used for statistical comparisons between dogs and between evaluation times. Changes in peak force, impulse, or slope and evaluation of the symmetry between limbs provide useful information about limb function. Changes in the distribution of force in the limbs caused by various types of lameness have also been evaluated.[17, 34, 35] In most research studies, force-plate data from dogs are pooled and compared with data from the same population at different times (i.e., before and after treatment).

Studies Using Force-Plate Analysis in Dogs

The force plate has been used to study normal gait and to evaluate experimentally induced and naturally occurring lameness in dogs. Numerous studies have been performed in healthy dogs to document normal gait and to identify sources of variation.[6, 9, 11, 20, 24, 25, 29–31, 32, 33, 38] These studies have led to a better understanding of normal canine gait as well as to improved data collection techniques. Other studies have evaluated the effects of lameness on force distribution to other limbs.[17, 34, 35] Additionally, force-plate gait analysis has been used to evaluate common causes of lameness and their various treatments, including hip dysplasia,[8, 10, 14, 22, 23, 26] cruciate ligament injury,[7, 15, 16, 28] fracture repair,[27] and osteoarthritis.[8, 37] The use of force-plate gait analysis in these studies provided accurate and repeatable data on limb function and an objective measurement of the efficacy of each treatment method. Currently, the force plate is primarily a research tool used to evaluate lameness and its treatment. As understanding of force-plate technology and experience with clinical lameness increase, its use as a diagnostic aid in clinical cases of lameness should increase.

KINEMATIC GAIT ANALYSIS

Collection of kinematic data allows accurate description of the complex movements of gait but requires specialized equipment.[1, 3, 12, 13, 18] Four video cameras linked to a video processor and computer are positioned to simultaneously record the subject's motion as it passes through the three-dimensional testing space. Retroreflective targets are placed on the dog's skin over specific anatomic landmarks. As the dog is walked or trotted through the testing space, the video cameras record the movement of the retroreflective targets. The computer then analyzes the data from all four cameras by direct linear transformation to determine the relative positions of each target in three-dimensional space.[12] With this technology, a great deal of information can be gathered, including subject velocity, the segmental velocities of each portion of the limb, stride length and frequency, joint angles (flexion and extension data), angular velocities, and temporal data.[1, 3, 12, 13, 18] This information can also be correlated with kinetic data by integration of a force plate into the testing space.[12]

Kinematic gait analysis of dogs has been used to study the normal gaits (walk and trot) and to evaluate dogs with hip dysplasia and cruciate ligament rupture.[1, 3, 12, 13, 18] Although this technology is relatively new to veterinary research, it holds great promise for future study of lameness and treatment of joint disease in dogs.

SUMMARY

Kinetic and kinematic gait analysis provides objective, quantifiable, and repeatable information on normal and abnormal gait in dogs. Data collection requires specialized equipment, and techniques must be carefully controlled to ensure that accurate measurements are obtained. Force-plate and kinematic analysis is currently used primarily as a research tool to study various gait abnormalities and objectively assess treatment efficacy. As future research identifies characteristic changes associated with specific types of lameness, the use of gait analysis to evaluate individual clinical patients with lameness should become more valuable. Specialized gait analysis techniques may eventually enable veterinarians to accurately diagnose subtle lameness, better evaluate dogs with resolving lameness, and accurately select the appropriate time to return an athletic dog to exercise after recovery from an injury.

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